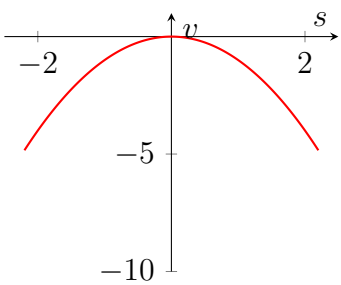
Marlos C. Machado
CMPUT 365: Introduction to Reinforcement Learning

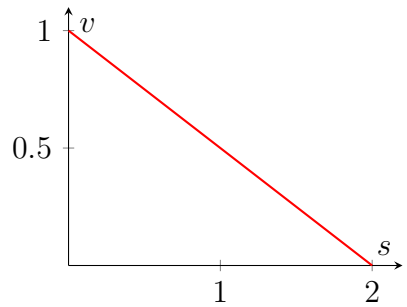
Worksheet 10: Constructing Features for Prediction

Question 1. Consider the following two functions.

$v(s) = -s^2$



$v(s) = 1.0 - \frac{1}{2}s$



(a) Design features for each function, to approximate them as a linear function of these features. Can you design features to make the approximation exact?

(b) Can you design one set of features that allows you to represent both functions?

(a) For the first function, we use a single feature: s^2. Using this feature, we can exactly (linearly) represent the function (with a weight of -1). For the second function, we use two features: a bias feature which is always 1 and s, i.e. we use the feature vector $(1, s)^\top$. Using these features, we can exactly represent the function with linear function using the weight vector $(1, -0.5)^\top$.

(b) Yes, the feature $(1, s, s^2)^\top$ can be used to exactly represent both the functions using linear function approximation.

Question 2. Consider a (one hidden layer) neural network, where the input *column* vector $\mathbf{s}$ is mapped by the input-weight matrix $\mathbf{B}$ and a (differentiable) activation function g to "feature vector"

$$\mathbf{x} \doteq g(\boldsymbol{\psi}), \qquad \text{where} \quad \boldsymbol{\psi} = \mathbf{B}\mathbf{s}.$$

(We assume that the function g is applied elementwise to the vector $\boldsymbol{\Psi}$, i.e. the above equation is equivalent to saying $x_i = g(\psi_i)$ for all i.) Finally, the feature vector is mapped by the output-weight vector $\mathbf{w}$ linearly to the value estimate

$$\hat{v} \doteq \mathbf{w}^\top \mathbf{x}.$$

(Here, we use c_i to denote the ith element of the vector $\mathbf{c}$, and use a_{ij} to denote the element at the ith row and the jth column of a matrix $\mathbf{A}$.) Now, using chain rule, we can compute the following derivatives:

$$\frac{\partial \hat{v}}{\partial w_j} = \frac{\partial}{\partial w_j}\left(\sum_i w_i x_i\right) = x_j,$$

$$\frac{\partial \hat{v}}{\partial B_{ij}} = w_i \frac{\partial x_i}{\partial B_{ij}} = w_i \frac{\partial g(\psi_i)}{\partial \psi_i}\frac{\partial \psi_i}{\partial B_{ij}} = w_i \frac{\partial g(\psi_i)}{\partial \psi_i} s_j,$$

where in the first equality on the second line, we used the fact that x_k for $k \neq i$ would be independent of B_{ij}.

For the following sub-questions, let the activation function be $ReLU$, which for $x \in \mathbb{R}$ is defined as:

$$f(x) = \begin{cases} x, & \text{if } x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) What is the derivative of the ReLU activation function g? When are the derivatives well-defined?

(b) Imagine that all the weights (i.e., $\mathbf{B}$ and $\mathbf{w}$) are initialized to zero. Would this be a problem? If so, why?

(a) The derivative for the ReLU activation g is as follows:

$$g'(\psi_i) = \begin{cases} 1, & \text{if } \psi_i > 0, \\ 0, & \text{if } \psi_i < 0, \end{cases}$$

and $g'(0)$ is not defined. However, we can define $g'(0)$ to be any value from $[0, 1]$. The g' defined in this way is a subgradient of g. Similar to gradient descent, we can use gradient and gradient descent to refer to subgradients and subgradient descent in the questions and solutions. In practice, we usually define $g'(0) = 0$ for some practical reasons (this is outside the scope of this question, see this post on Quora for more discussion[1]).

(b) Yes, this will be a major problem during the network training. If all the weights are zero, the derivative of the value estimate with respect to the weights will be zero as well:

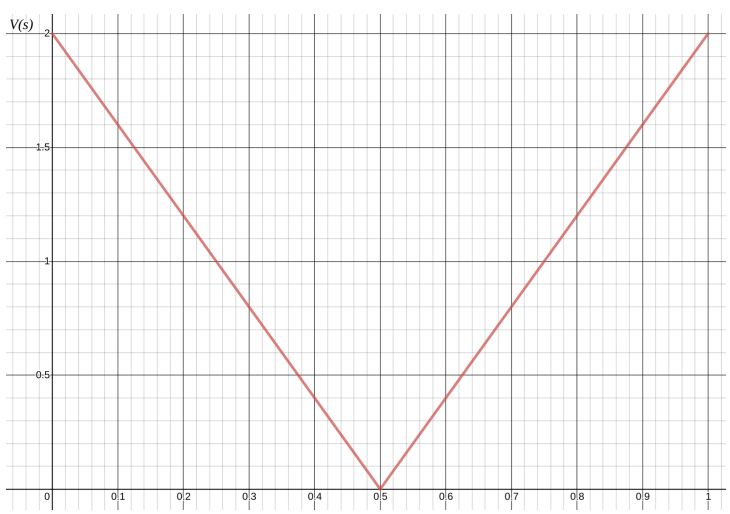
$$\begin{aligned} \mathbf{x} &= g(\mathbf{B}\mathbf{s}) = g(\mathbf{0}) = \mathbf{0}, && \text{(using } \mathbf{B} = \mathbf{0} \text{ and ReLU}(0) = 0) \\ \frac{\partial \hat{v}}{\partial w_j} &= \frac{\partial \mathbf{w}^\top \mathbf{x}}{\partial w_j} = x_j = 0, && \text{(using } \mathbf{x} = \mathbf{0}) \\ \frac{\partial \hat{v}}{\partial B_{ij}} &= w_i \frac{\partial x_i}{\partial B_{ij}} = 0, && \text{(using } \mathbf{w} = \mathbf{0}) \end{aligned}$$

Thus, any future updates to the system will not change its weights, and no learning would occur.

Question 3. Consider a problem with the state space, $\mathcal{S} = \{0, 0.01, 0.02, \cdots, 1\}$. Assume the true value function is

$$v_\pi(s) = 4|s - 0.5|,$$

which is visualized below.



[1] https://www.quora.com/How-do-we-compute-the-gradient-of-a-ReLU-for-backpropagation

Let us now approximate v_π using state aggregation. We choose to aggregate the states into two bins: $[0, 0.5]$ and $(0.5, 1]$. (This means that the value function learns, say, $w_1 \in \mathbb{R}$ for all $s \in [0, 0.5]$ and learns, say, $w_2 \in \mathbb{R}$ for all $s \in [0.5, 1]$. Also, let $\mathbf{w} = (w_1, w_2)^\top$.) Then,

(a) give one set of possible features $\mathbf{x}$, such that a linear approximation to v_π (i.e. $v_{\mathbf{w}}(s) = \mathbf{w}^\top \mathbf{x}(s)$) is equivalent to the value estimate learned using the (two bin) state aggregation technique described above.

(b) Imagine you minimize the (weighted) mean squared value error

$$\overline{\mathrm{VE}}(\mathbf{w}) \doteq \sum_{s \in \mathcal{S}} d(s)(v_\pi(s) - \hat{v}(s, \mathbf{w}))^2,$$

with a uniform weighting $d(s) = \frac{1}{101}$ for all $s \in \mathcal{S}$. Here $\hat{v}$ represents the value function estimate under the state-aggregation scheme described before. Then compute

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \in \mathbb{R}^2} \overline{\mathrm{VE}}(\mathbf{w}).$$

(c) Now, if the agent puts all of the weighting on the range $[0, 0.25]$, (i.e. $d(s) = 0$ for all $s \in (0.25, 1]$ and $d(s)$ is uniform otherwise), then what vector $\mathbf{w}^*$ is found by minimizing $\overline{\mathrm{VE}}$?

(a) One possible set of features could be

$$\mathbf{x}(s) = (0, 1)^\top \text{ for all } s \in [0, 0.5], \quad \text{and} \quad \mathbf{x}(s) = (1, 0)^\top \text{ for all } s \in [0.5, 1].$$

(b) Note that

$$v_\pi(s) = \begin{cases} -4s + 2, & \text{if } s \in [0, 0.5], \\ +4s - 2, & \text{if } s \in (0.5, 1]. \end{cases}$$

Then,

$$\overline{\mathrm{VE}}(\mathbf{w}) = \frac{1}{101} \left[\sum_{s \in \{0, 0.01, \ldots, 0.5\}} (-4s + 2 - w_1)^2 + \sum_{s \in \{0.51, 0.52, \ldots, 1\}} (4s - 2 - w_2)^2 \right].$$

Clearly, we can minimize the above two sums separately. Doing so, we obtain:

$$w_1^* = \frac{1}{51} \sum_{s \in \{0, 0.01, \ldots, 0.5\}} (-4s + 2) = 2 - \frac{4}{51}(0 + 0.01 + \cdots + 0.5) = 2 - \frac{4}{51} \cdot \frac{0.5 \cdot 51}{2} = 1,$$

$$w_2^* = \frac{1}{50} \sum_{s \in \{0.51, 0.52, \ldots, 1\}} (4s - 2) = \frac{4}{50}(0.51 + 0.52 + \cdots + 1) - 2 = \frac{4}{50} \cdot \frac{1.51 \cdot 50}{2} - 2 = 1.02.$$

(c) For this weighting $d(\cdot)$, we get

$$\overline{\mathrm{VE}}(\mathbf{w}) = \frac{1}{26} \sum_{s \in \{0, 0.01, \ldots, 0.25\}} (-4s + 2 - w_1)^2.$$

Clearly, w_2^* can be anything. Minimizing the above expression with respect to w_1, we get

$$w_1^* = \frac{1}{26} \sum_{s \in \{0, 0.01, \ldots, 0.25\}} (-4s + 2) = 2 - \frac{4}{26}(0 + 0.01 + \cdots + 0.25) = 2 - \frac{4}{26} \cdot \frac{0.25 \cdot 26}{2} = 1.5.$$

Question 4. Consider the following general SGD update rule with a general target U_t

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha \left[U_t - \hat{v}(S_t, \mathbf{w}_t)\right] \nabla \hat{v}(S_t, \mathbf{w_t}).$$

Assume that we are using linear function approximation, *i.e.* $\hat{v}(S, \mathbf{w}) = \mathbf{x}(S)^\top \mathbf{w}$.

(a) What happens to the update if we scale the features by a constant? In particular, we use the new features $\tilde{\mathbf{x}}(S) = 2\mathbf{x}(S)$. Will this pose any problem? Why or why not?

(b) In general, we want a stepsize that is invariant to the magnitude of the feature vector $\mathbf{x}(S)$, where the magnitude is measured by the inner product $\mathbf{x}(S)^\top \mathbf{x}(S)$. The book suggests the following stepsize:

$$\alpha = \frac{1}{\tau \mathbf{x}(S)^\top \mathbf{x}(S)}.$$

What is $\mathbf{x}(S)^\top \mathbf{x}(S)$ when using tile coding with 10 tilings? Suppose $\tau = 1000$, what is α if we use tile coding with 10 tilings?

(a) Given the features $\mathbf{x}$, the update is

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha[U_t - \mathbf{x}(s)^\top \mathbf{w}_t]\mathbf{x}(s).$$

With the new features $\tilde{\mathbf{x}}$, the update becomes

$$\mathbf{w}'_{t+1} = \mathbf{w}'_t + 2\alpha[U_t - 2\mathbf{x}(s)^\top \mathbf{w}'_t]\mathbf{x}(s).$$

In general, if we control for the stepsize (see the next sub-part), this would not create any problems. And we expect that the final weight vector $\mathbf{w}'_\infty$ learned in the latter case would be half as big as the weight vector $\mathbf{w}_\infty$ in the former case.

(b) Since there are 10 tilings, $\mathbf{x}(S)$ would be a binary feature vector (length of which depends on size of each tiling), with exactly 10 unit elements, and the rest being zero. Therefore, $\mathbf{x}(S)^\top \mathbf{x}(S) = 10$ and $\alpha = \frac{1}{1000 \cdot 10} = 10^{-4}$.

Question 5. A student is tasked to program an agent for a 100×100 grid-world problem, with the states denoted by (x, y) and $x, y \in \{0, 1, \ldots, 99\}$. The goal of this problem is to reach the state $(99, 99)$, located at the top-right corner. In this grid-world problem, rewards are -1 at each time step, except for the terminal state where the reward is 0 and the episode ends. The student first considers a tabular feature representation. However, since the grid-world has a large number of states, the student is worried about generalization.

(a) What is the dimension of the tabular representation? Suppose the student chooses to represent a state by its (x, y) coordinates instead. What is the dimension of this new representation? How does it differ from the tabular representation?

(b) Design a function $f(x, y)$ as features for linear Temporal Difference (TD) learning that, where (x, y) are the state coordinates, can improve the agent's performance in this fixed grid-world problem.

(c) Suppose that the student designs a new feature representation $\mathbf{x}'(s) = (x, y, z)$, where the first two components are the row and column indices, and the last component is binary: $z = 1$ when the agent is 2 squares away from the goal and $z = 0$ otherwise. What is the dimension of feature vector in this case? How much will this help with generalization instead of only using $\mathbf{x}(s) = (x, y)$?

(a) The dimensionality of the tabular representation is: $100 \times 100 = 104$. The dimensionality of the second representation is 2. This new representation would generalize across the X and Y axes respectively, whereas the tabular representation has no generalization.

(b) One possible feature is the Manhattan distance to the goal. $f(x, y) = (99 - x) + (99 - y)$.

(c) The feature vector is 3 dimensional. Further, it is correlated with the goal state, and provides more fine-grained distinction between states that are close to the goal state and those which are not.

Question 6. Exercise 9.3 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 7. Exercise 9.4 from Sutton & Barto (2018).

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Question 8. Exercise 9.6 from Sutton & Barto (2018).

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